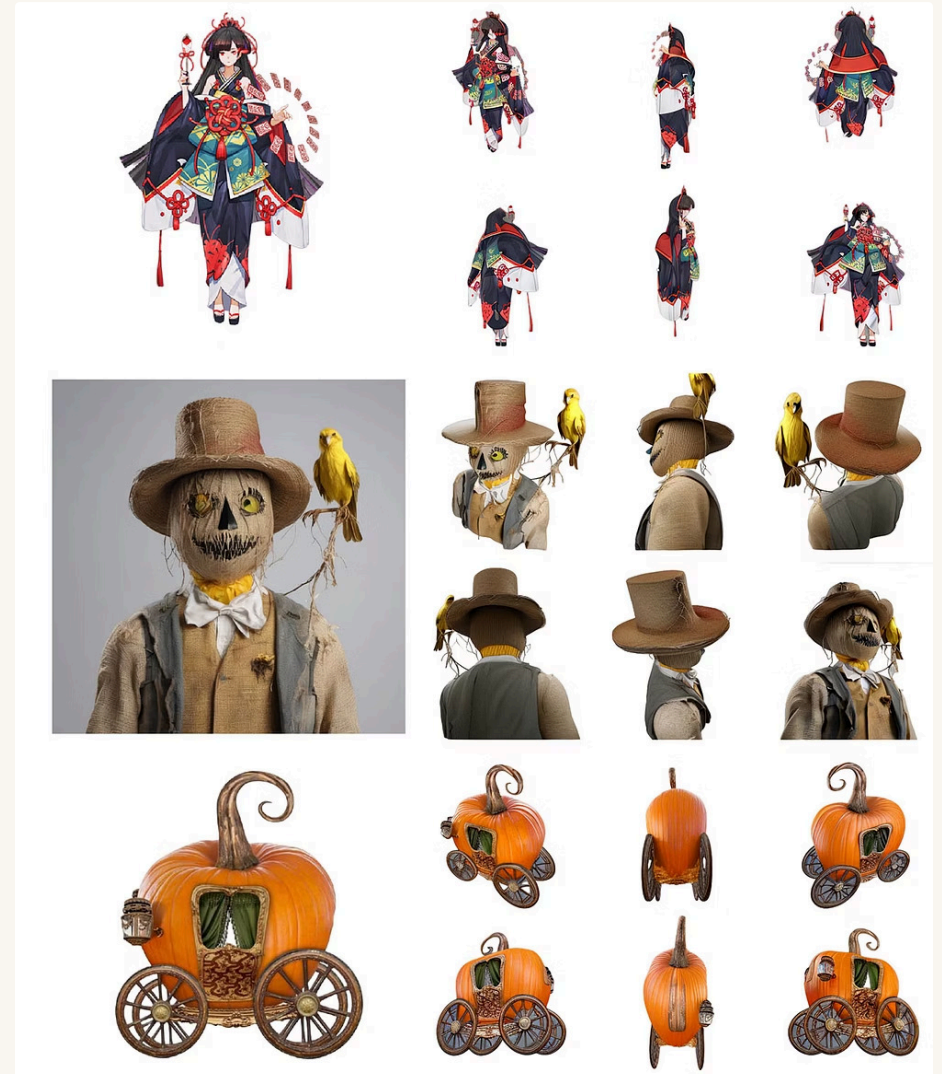


Zero123++: a Single Image to Consistent Multi- view Diffusion Base Model

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Zero123++ model:

- What is it? Zero123++ is an image-conditioned diffusion model.
- What does it do? It generates **3D-consistent, multi-view images** from a **single input view**.
- How? It's finetuned from an off-the-shelf model (Stable Diffusion) using new conditioning and training schemes to leverage existing 2D priors.
- Why is it better? It excels at producing high-quality, consistent images, overcoming common issues like **texture degradation** and **geometric misalignment**.
- Bonus Feature: The authors also demonstrate a **ControlNet** trained on Zero123++ for enhanced control over the generation process.

Some previous works that led to Zero123++

- The foundation: **Stable Diffusion**
- The Direct Predecessor: **Zero-1-to-3**
- Concurrent Consistency Models: **SyncDreamer**, **Consistent123**
- Downstream 3D Reconstruction Models: **One-2-3-45**, **DreamGaussian**

Concurrent Consistency Models

Other models that also tried to solve the consistency problem of Zero-1-to-3. For instance, the paper cites:

- **SyncDreamer:** Uses a 3D-aware attention mechanism to synchronize the generation of multiple views at once.
- **Consistent123:** Another model that builds "extra layers" on Zero-1-to-3 to get more 3D-consistent results.

Downstream 3D Reconstruction Models

Models that *consume* multi-view images (like those from Zero123) to create the final 3D asset (e.g., a mesh or Gaussian Splat). For instance, the paper cites:

- **One-2-3-45:** A feed-forward model that can turn a single image into a 3D mesh in seconds, using a multi-view diffusion model (like Zero123) as its first step.
- **DreamGaussian:** An optimization-based method that uses a 2D diffusion prior to create a 3D Gaussian Splatting representation, which is much faster than traditional NeRFs.

Outline

1. Stable Diffusion

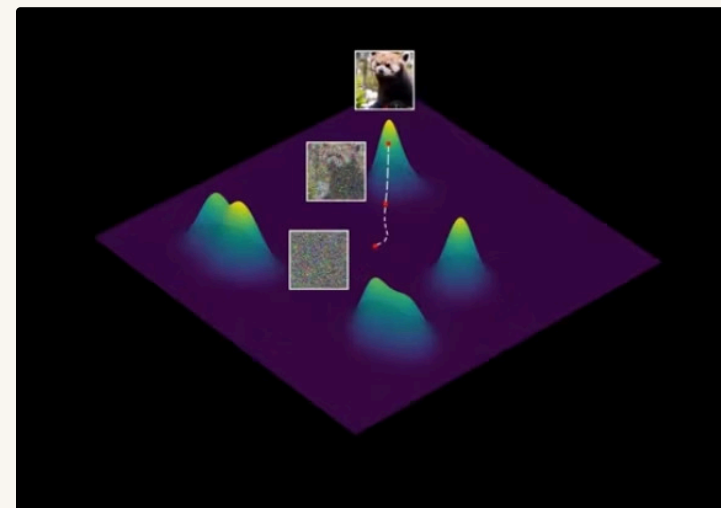
- a. Recall of a Diffusion Model
- b. A Latent Space Models
- c. CLIP
- d. U-Net

2. Zero123++ improvements

- a. Multi-view Generation
- b. Consistency and Stability: Noise Schedule
- c. Local Condition: Scaled Reference Attention
- d. Global Condition: FlexDiffuse
- e. Extension: Depth-Controlled Zero123++

I. Stable Diffusion

- **What it is:** A foundational, open-source 2D text-to-image diffusion model.
- **Relevance:** Zero123++ is a **direct finetune** of the **Stable Diffusion 2 v-prediction model**. The paper's core philosophy is to re-use and adapt its powerful 2D priors, not start from scratch.
- **To build it :**
 - Latent Space Models
 - CLIP
 - U-Net



a) Recall : Diffusion Model

1. Forward Process (Adding Noise)

The noisy image x_t is created from the clean image x_0 and the noise ϵ .

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon \quad \text{where} \quad \epsilon \sim \mathcal{N}(0, I)$$

With $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$ and $\alpha_t = 1 - \beta_t$

2. Noise Schedule

a) Linear Schedule

β_t (the noise added at each step) **increases linearly** from a starting value to an ending value.

$$\beta_t = \beta_{start} + \frac{t}{T}(\beta_{end} - \beta_{start})$$

b) Scaled-Linear Schedule

The goal is to have a smoother $\bar{\alpha}_t$ (cumulative signal) curve and to better manage the **Signal-to-Noise Ratio (SNR)**

$$\beta_t = \left(\sqrt{\beta_{start}} + \frac{t}{T}(\sqrt{\beta_{end}} - \sqrt{\beta_{start}}) \right)^2$$

3. Reverse Process (Denoising by U-Net)

a) Epsilon-Parameterization (ϵ -prediction)

The U-Net, denoted $\epsilon_\theta(x_t, t)$, is trained to **directly predict the noise** ϵ that was used to create x_t .

- **Prediction Target:** ϵ_t
- **Loss Function (simplified):** $L = ||\epsilon_t - \epsilon_\theta(x_t, t)||^2$
- **How to find x_0 :** $x_0^{pred} = \frac{1}{\sqrt{\bar{\alpha}_t}}(x_t - \sqrt{1 - \bar{\alpha}_t}\epsilon_\theta(x_t, t))$

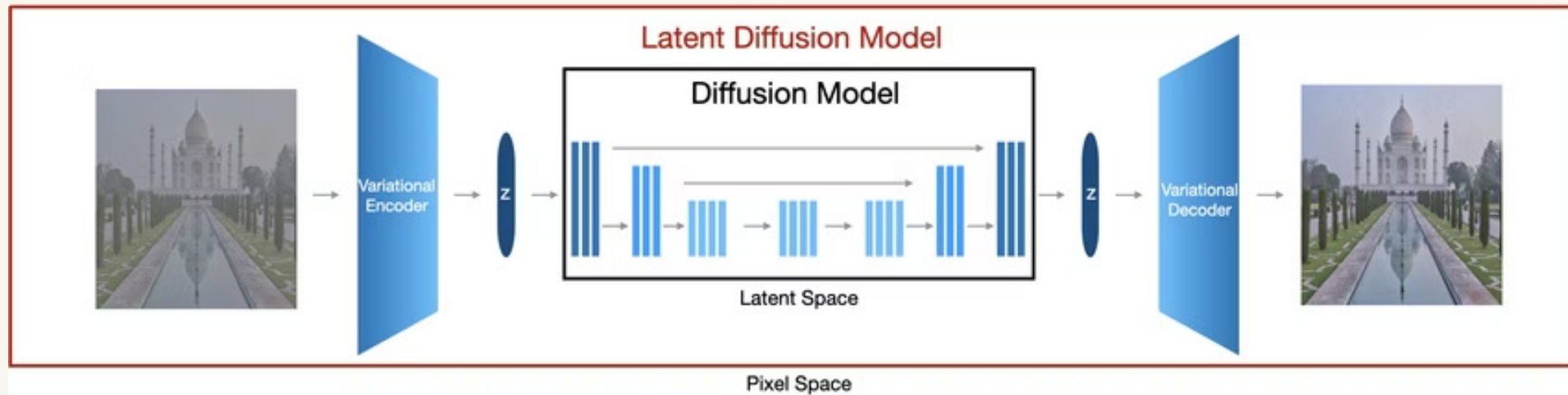
b) Parametrization-V (v -prediction)

The U-Net, denoted $v_\theta(x_t, t)$, is trained to predict a **stable target** v (the "velocity").

- **Prediction Target:** v_t
- **Loss Function (simplified):** $L = ||v_t - v_\theta(x_t, t)||^2$
- **How to find x_0 :** $x_0^{pred} = \sqrt{\bar{\alpha}_t}x_t - \sqrt{1 - \bar{\alpha}_t}v_\theta(x_t, t)$

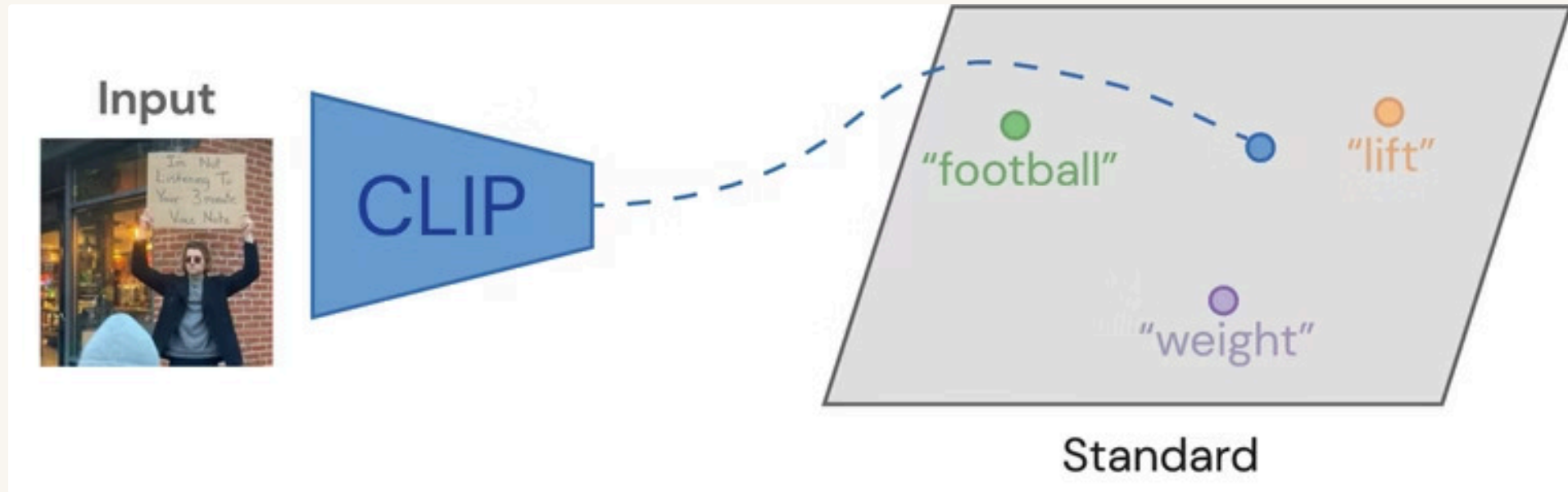
b) A Latent Space Models

Unlike traditional diffusion models that operate directly on pixel space, Stable Diffusion uses a latent space representation, reducing computational requirements and allowing efficient generation of high-resolution images.



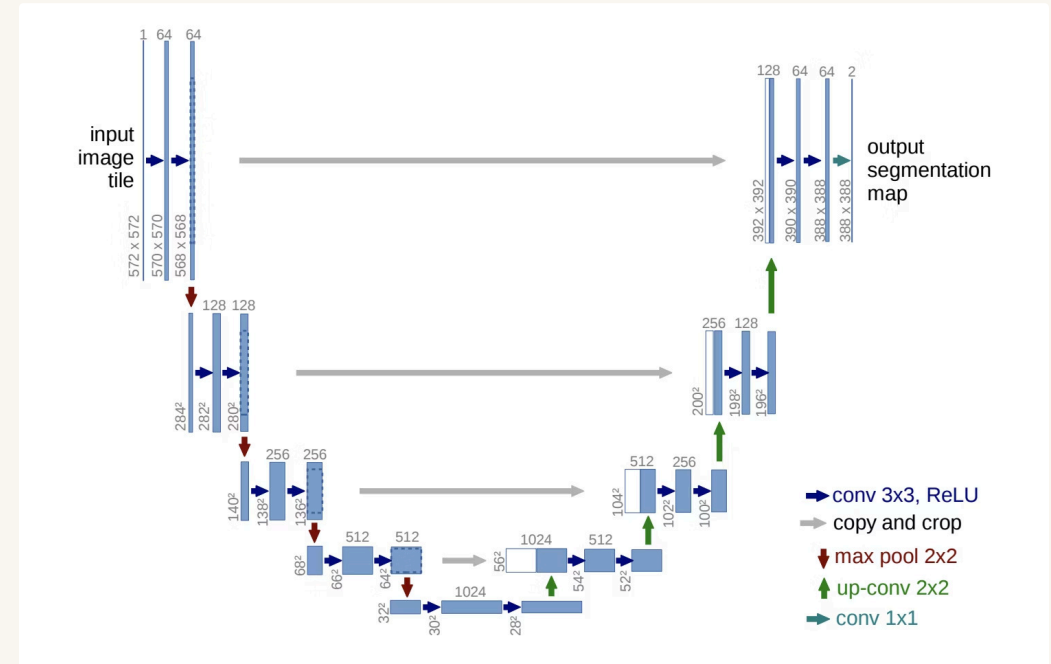
c) CLIP (Contrastive Language–Image Pre-training).

- **Text Encoder** (Transformer): Converts text into a vector.
- **Image Encoder** (ViT/ResNet): Converts the image into a vector.
- A **cosine distance** is then used to estimate how close the text and the image are.



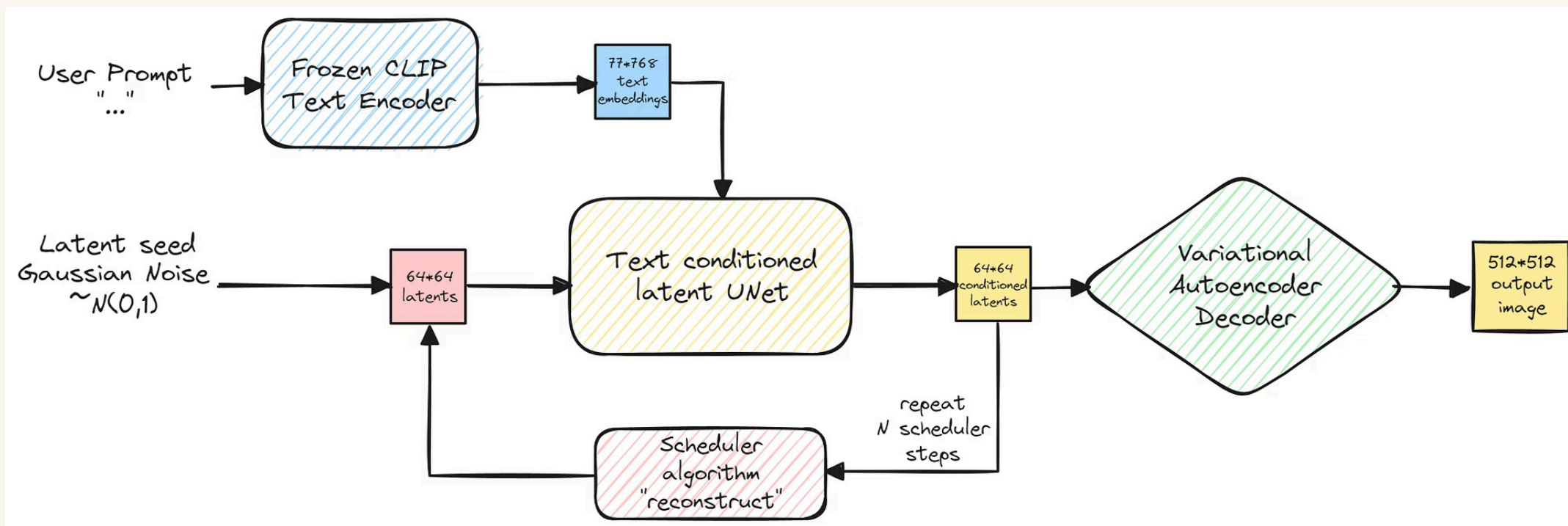
d) U-Net Architecture

- Combining a **contracting path** (resolution reduction) and an **expanding path** (resolution increase).
- Skip Connections:** Direct connections link layers from the contracting path to symmetric layers in the expanding path to preserve fine-grained **localization information**.



Stable Diffusion Model

The text is incorporated into the U-Net through a mechanism called Cross-Attention.



II. Zero123++ improvements

- **Zero-1-to-3:**
 - Original model that pioneered zero-shot "one image to 3D object" by finetuning Stable Diffusion
 - Two problems:
 - Independent View Generation
 - Underutilizing Stable Diffusion
- **Zero123++ improvements:**
 - Multi-view Generation
 - Consistency and Stability: Noise Schedule
 - Local Condition: Scaled Reference Attention
 - Global Condition: FlexDiffuse

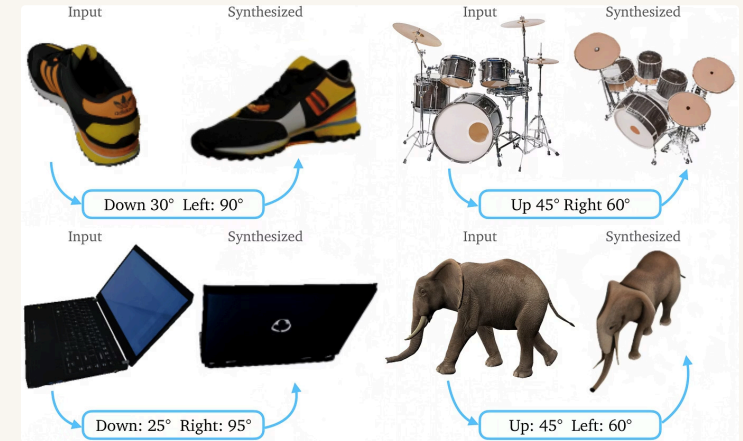


Image from the website "Zero-1-to-3: Zero-shot One Image to 3D Object"

a) Multi-View Generation

1. The Problem: Lacking Consistency

- **Zero-1-to-3** generates each new view **independently**.
- **Consequence:** "breakdown in consistency" between the generated views (e.g., textures don't match, parts disappear).

2. The Solution: Tiling

- Zero123++ 's new strategy: **tiling all six views into a single 3x2 image**.

3. The Benefit: Modeling Joint Distribution

- Forces the model to **generate all views at once**.
- Learns the **joint distribution** of the multi-view images, ensuring consistency.



Modeling Joint Distribution

The Goal: Model the Joint Distribution. The model must learn the **joint probability distribution** of all 6 views ($V_1 \dots V_6$) given the input image I_{cond} .

$$p(V_1, V_2, V_3, V_4, V_5, V_6 \mid I_{cond})$$

The Problem: Zero-1-to-3's Independent Approach

- For each view learns: $p(V_n \mid I_{cond})$.
- *Implicitly* assumes:

$$p(V_1, \dots, V_6 \mid I_{cond}) \approx \prod_{n=1}^6 p(V_n \mid I_{cond})$$

- **Result:** ignores the correlations between multi-view images, leading a consistency breakdown.

The Solution: Zero123++ Tiling Strategy

- Zero123++ **tiles all 6 views into a single 3×2 frame** and learns to generate this **single tiled image** conditioned on the input:

$$p(I_{tile} \mid I_{cond})$$

- **Result:** This forces the model to learn the **correct joint distribution** and the correlations between all views at once.

The Camera Pose Challenge

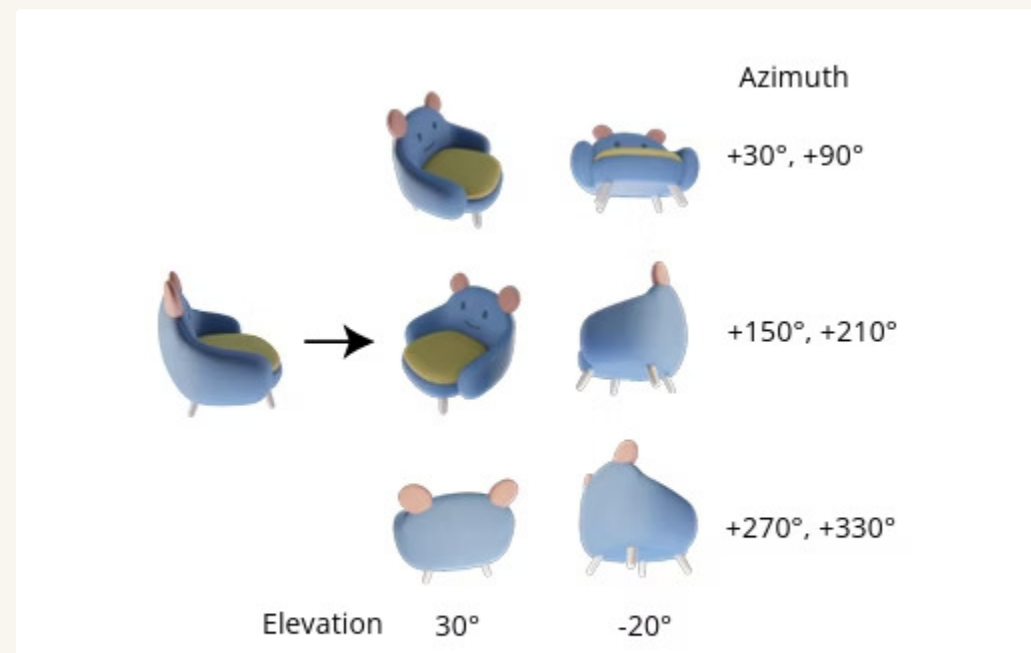
The Problem: How to define the camera angles for the 6 views?

- **Approach 1: Absolute Poses**
 - Issue is **Objaverse dataset** has no consistent "canonical pose".
 - Training on this leads to "**difficulties in disambiguating object orientations**".
- **Approach 2: Relative Poses (Used by Zero-1-to-3)**
 - Requires **knowing the elevation angle of the input view** to work.
 - Forces other pipelines (like One-2-3-45) to add an "elevation estimation module," which "**introduces extra error**".

The Camera Pose Solution

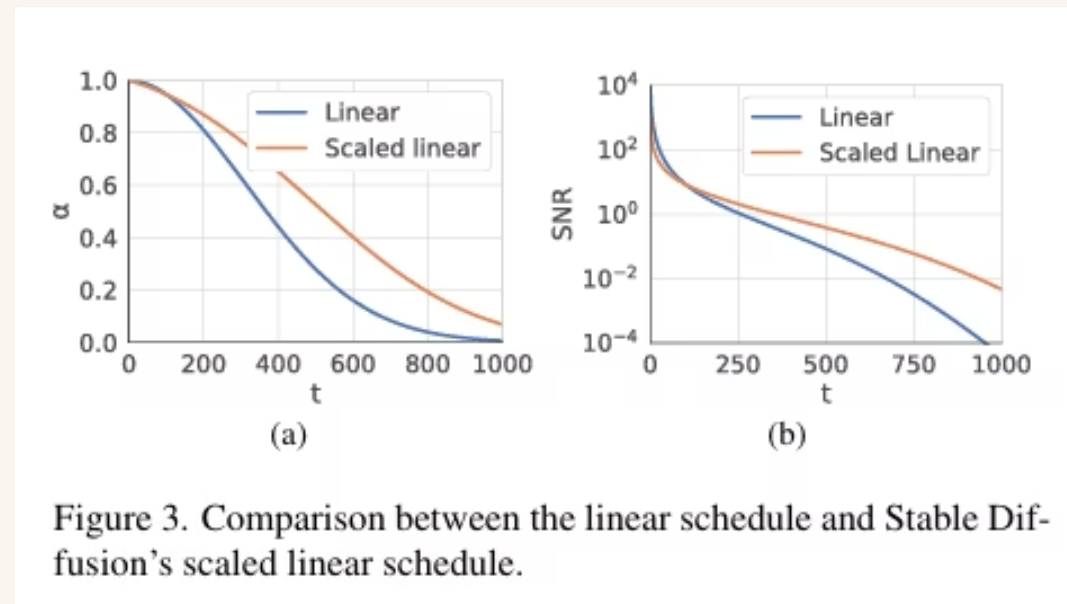
The Zero123++ Solution: A Hybrid Approach The model uses a *fixed* set of poses that **eliminates ambiguity** *without* needing an elevation estimator.

- **Fixed Absolute Elevation:** Uses interleaving angles of 30° downward and 20° upward.
- **Relative Azimuth:** Starts at 30° and increases by 60° for each of the 6 poses.



b) Consistency and Stability: Noise Schedule

- Stable Diffusion's default "**scaled-linear**" schedule is optimized for **local details**.
- Very few steps at **low Signal-to-Noise Ratio (SNR)**, which are crucial for learning **global structure**.
- Fine for single images, but limits **global consistency**.

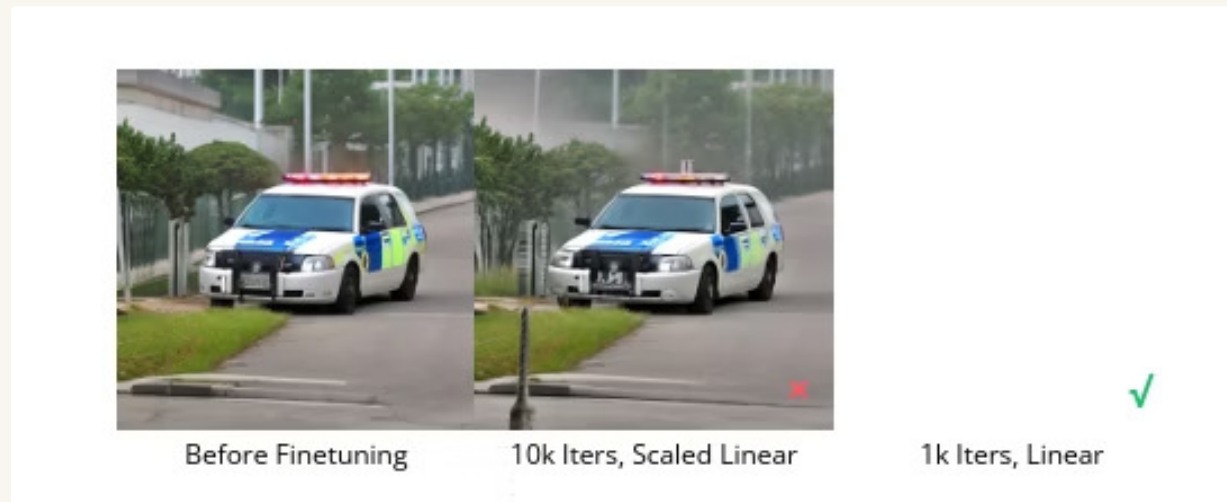


As seen in Figure 3: The "Scaled Linear" (orange) curve stays at a high SNR (low noise) for much longer than the "Linear" (blue) curve. The linear schedule forces the model to work on global structure (low-SNR) much earlier and for longer.

Proof: A Toy "Overfitting" Task

To test this, the authors designed a "toy task" to see how well each schedule could adapt to a new **global requirement**.

- **The Goal:** Finetune a LoRA model to output a **blank white image** for the prompt "a police car".
- **The Results :**



- **Scaled-Linear Schedule (SD Default): Failed.** The model "cannot overfit". It was "stuck" on its local-detail priors and only "slightly whitened the image".
- **Linear Schedule (New Choice): Success.** The model "successfully generates a blank white image", proving it could learn the new global requirement.

The Insight: Why Low-Res Training Was Unstable

This analysis also explains a major problem from the original **Zero-1-to-3** model:

- **The Clue:** Zero-1-to-3 trained at **low 256px resolution** because training at **512px was unstable**.
- **The Reason:** **More pixel redundancy** in high resolution, **appear less noisy** when **same amount of noise is added**.
- **The insight:** **Training at 256px** inadvertently prioritized global structure, effectively **mimicking a different noise schedule**.
- **Conclusion:** The instability happened because the original model was fighting its own (scaled-linear) schedule, which was not designed for this kind of global task at high resolution.

The Solution: Linear Schedule + v-Prediction

- **Step 1: Switch the Schedule.** Authors concluded it was necessary to switch to the **linear schedule** to force the model to learn global consistency.
- **Step 2: Solve the New Problem.** Swapping the schedule on a pretrained model is dangerous and can break it.
- **Step 3: The Key Discovery (Fig. 5).** The **v-prediction model** (from Stable Diffusion 2) is "**quite robust**" to this schedule swap. Other models (like the standard ϵ -prediction) are not.
- **Final Choice:** Use the **Stable Diffusion 2 v-prediction model**. This provides the *only* stable path to combine powerful SD priors with the new linear schedule needed for 3D consistency.



c) Local Condition: Scaled Reference Attention

Goal: How to give the model the *texture* and *local details* from the input image.

- **Zero-1-to-3's Method: Feature Concatenation.** The input image's features were just "stitched" onto the noisy latents.
- **The Problem:** Imposes an **incorrect pixel-wise spatial correspondence**.
 - *I.e.* tries to map pixels from the *front view* directly onto the *side view*, which doesn't make sense.
- **Zero123++'s Solution:** Use **Reference Attention**, a more flexible method to provide proper local conditioning.

How Reference Attention Works

Reference Attention allows the new view (the "Query") to "look up" details from the input image (the "Reference").

1. **Dual Branches:** The UNet runs on the *new view* (Noisy Latents) and the *input image* (Condition Image Latents) in parallel.
2. **Append K & V:** In the self-attention layers, the **Key (K)** and **Value (V)** matrices from the input image are **appended** to the K and V matrices of the new view.
3. **Add Same Noise:** The **same level of noise** (from timestep t) is added to *both* images.
 - **Why?** This lets the model find relevant features at the *current* stage of denoising.

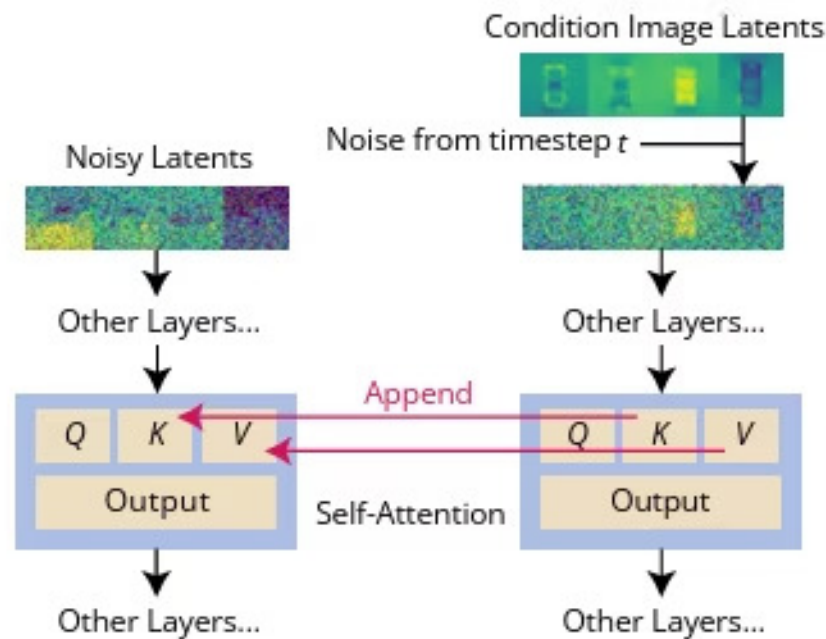
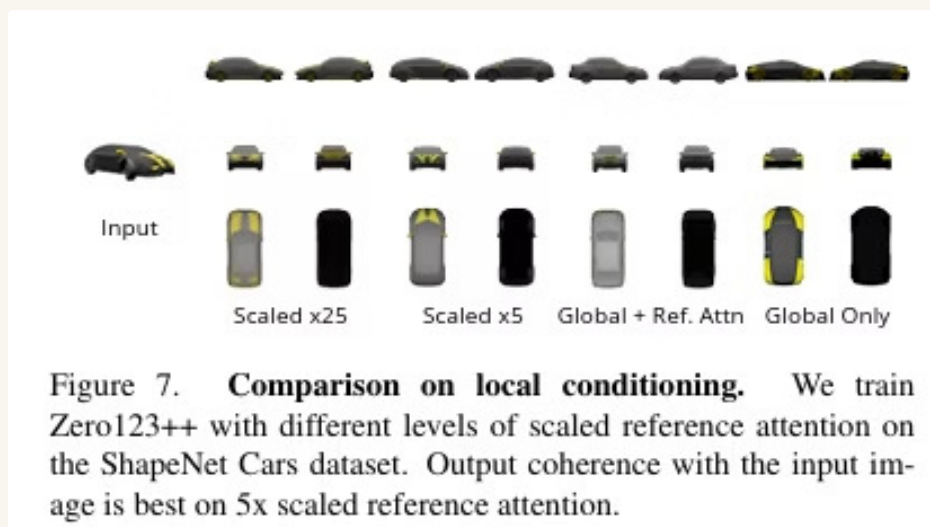


Figure 6. **Reference Attention.** It adds an additional conditioning branch and modifies key (K) and value (V) matrices of the self-attention layers to accept the extra condition image, which can fully reuse Stable Diffusion priors.

The Improvement: "Scaled" Reference Attention (Fig. 7)

- **The Tweak:** Authors found that Reference Attention works better during finetuning if they "**scale the latent (before adding noise)**".
- **The Experiment:** Tested different scaling factors on the ShapeNet Cars dataset.
- **The Result:** Highest consistency when reference latent was scaled by factor of 5.



d) Global Condition: FlexDiffuse

- **The Goal:** Inject global semantic context (e.g., "this is a car") without text prompts.
- **The Method:** Reuse Stable Diffusion's cross-attention by injecting the **CLIP Image Embedding** I into the empty text prompt T .
- **The Mechanism:** Add global image embedding to prompt tokens using linearly increasing weights w_i .

$$T'_i = T_i + w_i \cdot I, \quad \text{where } w_i = \frac{i}{L}$$

- **Initialization:** Weights initialized via **FlexDiffuse** linear guidance (i/L) to minimize finetuning effort.
- **The Impact:** Critical for generating **unseen regions** (e.g., the back of an object); prevents semantic degradation where local cues are missing.



e) Extension: Depth-Controlled Zero123++

1. The Mechanism: ControlNet

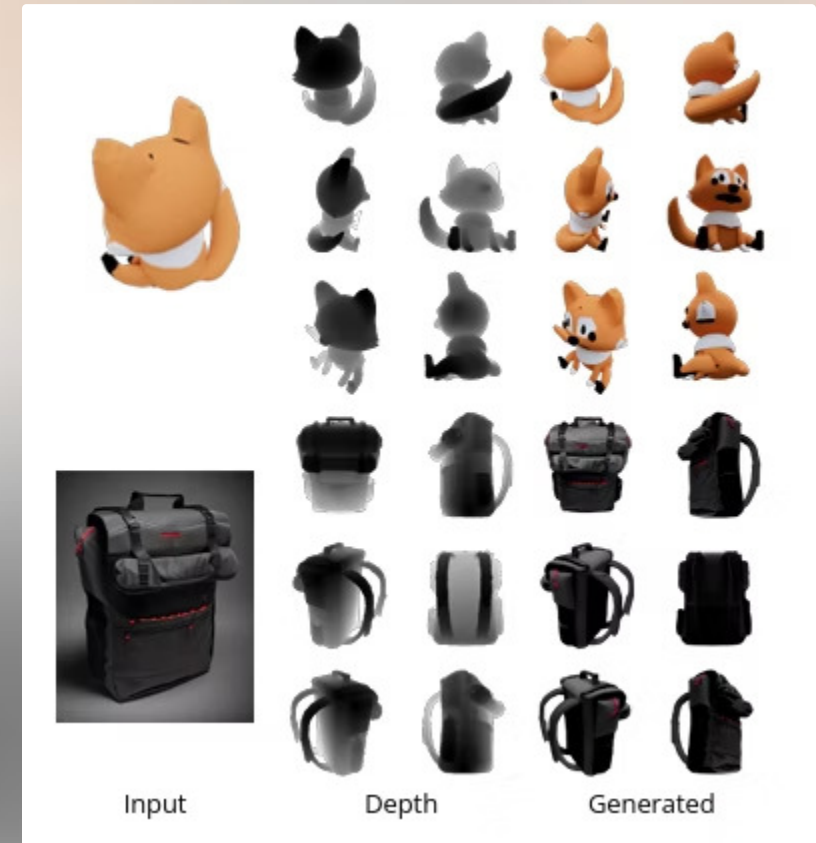
- The authors released a version of Zero123++ integrated with **ControlNet1**.
- **Training:** trained on normalized linear depth images paired with the target RGB images.

2. The Performance

- **Superior Consistency:** achieves better LPIPS score (**0.086**) than base model (**0.177**) on validation split.
- Proves that adding geometric guidance significantly stabilizes the output.

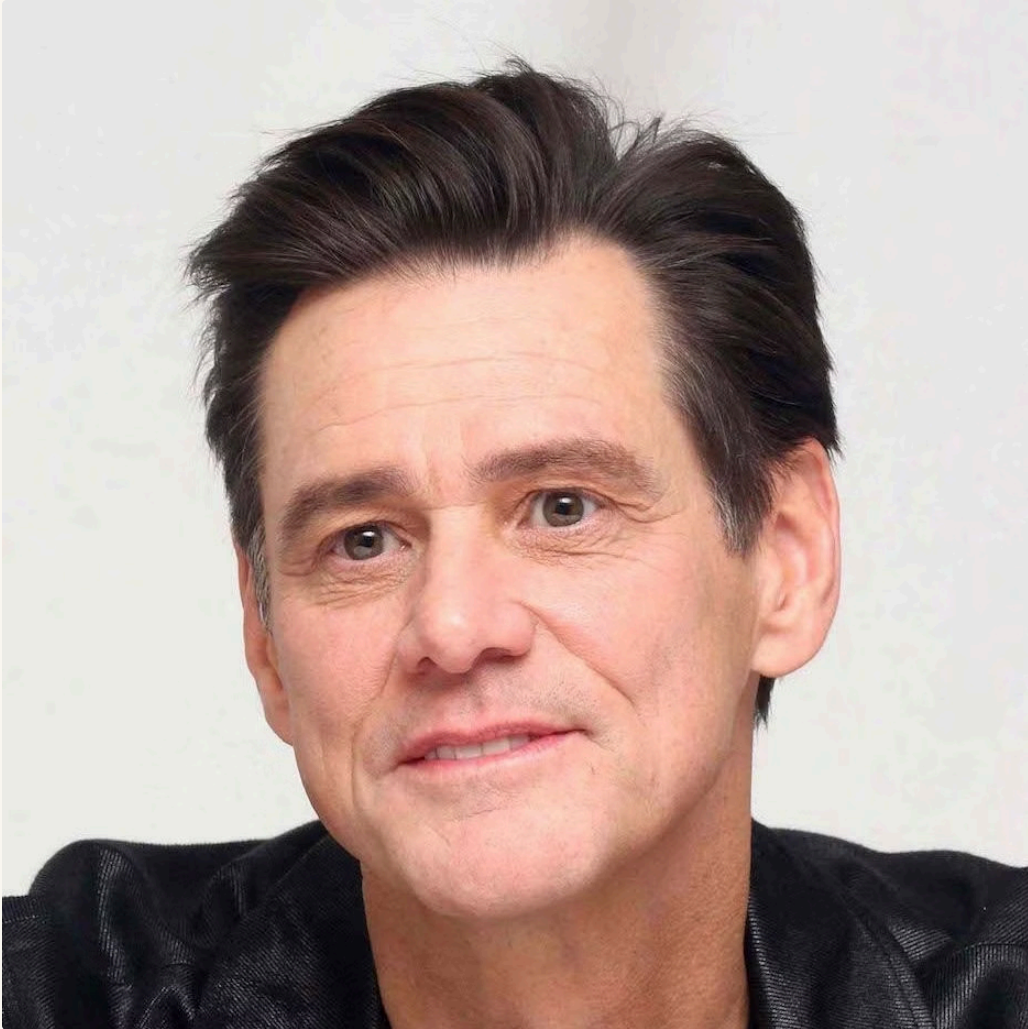
3. Key Use Case: Geometry-to-Appearance

- Allows for a "**Geometry** → **Multi-view**" pipeline.
- You can start with *only* a rough 3D shape (no colors), render its depth, and use the model to "paint" consistent textures onto it from all angles.



Results and critics







Future Works

- Use a new/bigger Dataset
 - They want to use **Objaverse-XL**
 - Maybe human datasets
- Go back to ϵ -parametrization
- Define clear 3D reconstruction pipeline
 - 3D Gaussian Splatting → **GaussianObject: High-Quality 3D Object Reconstruction from Four Views with Gaussian Splatting** ([Chen Yang](#), [Sikuang Li](#), [Jiemin Fang](#), [Ruofan Liang](#), [Lingxi Xie](#), [Xiaopeng Zhang](#), [Wei Shen](#), [Qi Tian](#))

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